TUM School of Life Science
Working for One Heath

Plant
Genetics

Omics Data Analysis of Root-Microbe Interactions

Lecturer Assistant
Tian Tian

Main content

- I Omics Data in Root – Microbe Studies (May 20)
- II Key Statistical Metrics in Microbiome Analysis (May 27)
- III Root transcriptome data analysis (June 3)
- IV Gene Expression – Microbiome Co-regulation Networks (June 10)

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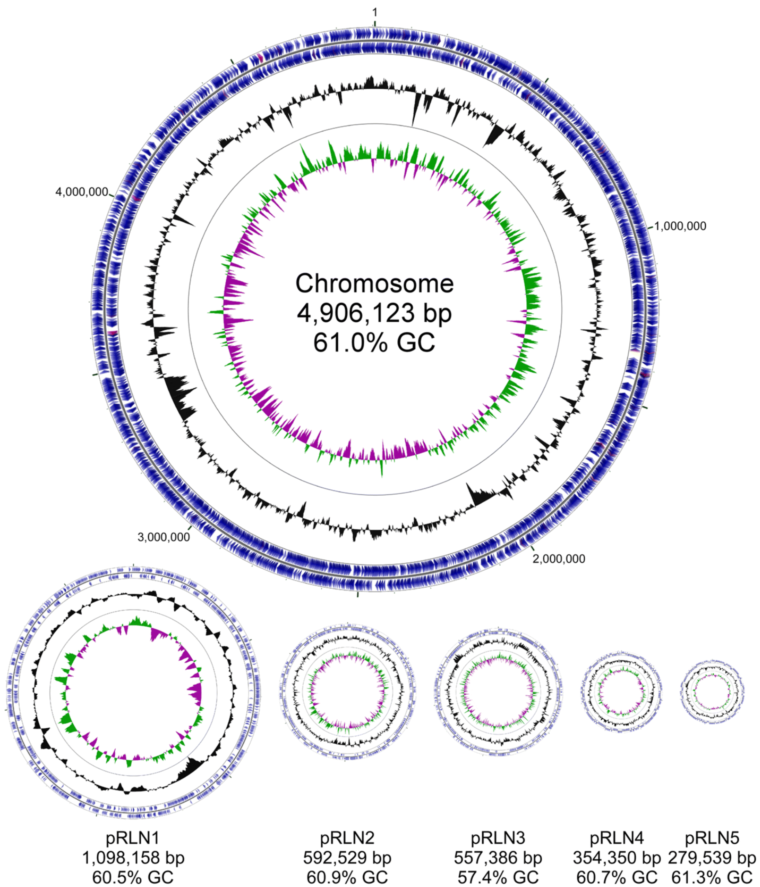
I Omics Data in Root – Microbe Studies (20.5)

Omics Data in Root–Microbe Studies

- **Research objective:** To understand the main types of omics data relevant to microbiome research.

- **16S rRNA Gene Sequencing** for bacterial/archaeal profiling
- **ITS Sequencing** (Internal Transcribbed Spacer) for fungal community profiling
- **Whole-Genome Sequencing** for comprehensive genetic analysis
- **Transcriptome Sequencing** (RNA-seq) for gene expression analysis

Bacterial genome



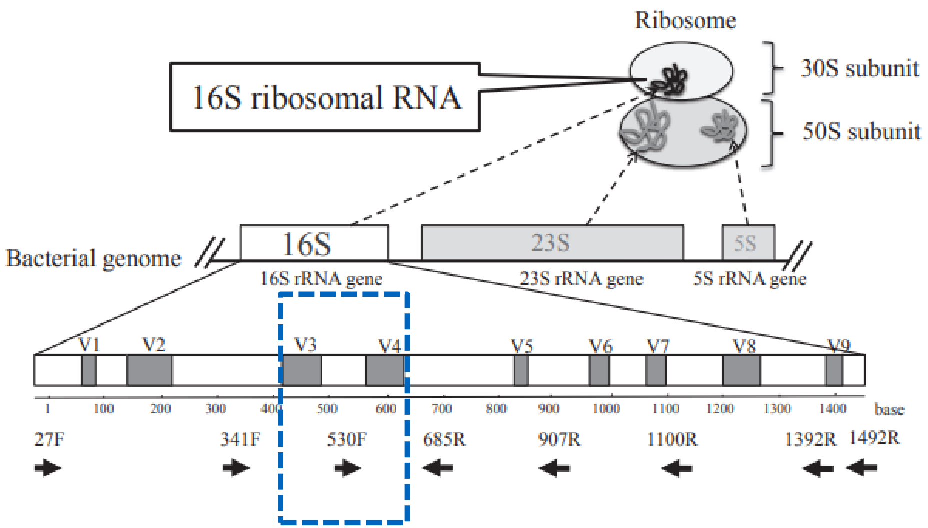
The chromosome and five plasmids of Rl Norway.

(Liang et al., 2018, Environmental Microbiome)

- Key Features of a Bacterial Genome:
  - Structure: Single circular chromosome + multiple small plasmids.
  - Size: Highly variable (0.5–10 Mb).
  - Assembly: Requires strain-level sequencing.
  - Classification: 16S rRNA gene as key marker.

Explore bacterial genomes:
MaGe platform: https://mage.genoscope.cns.fr

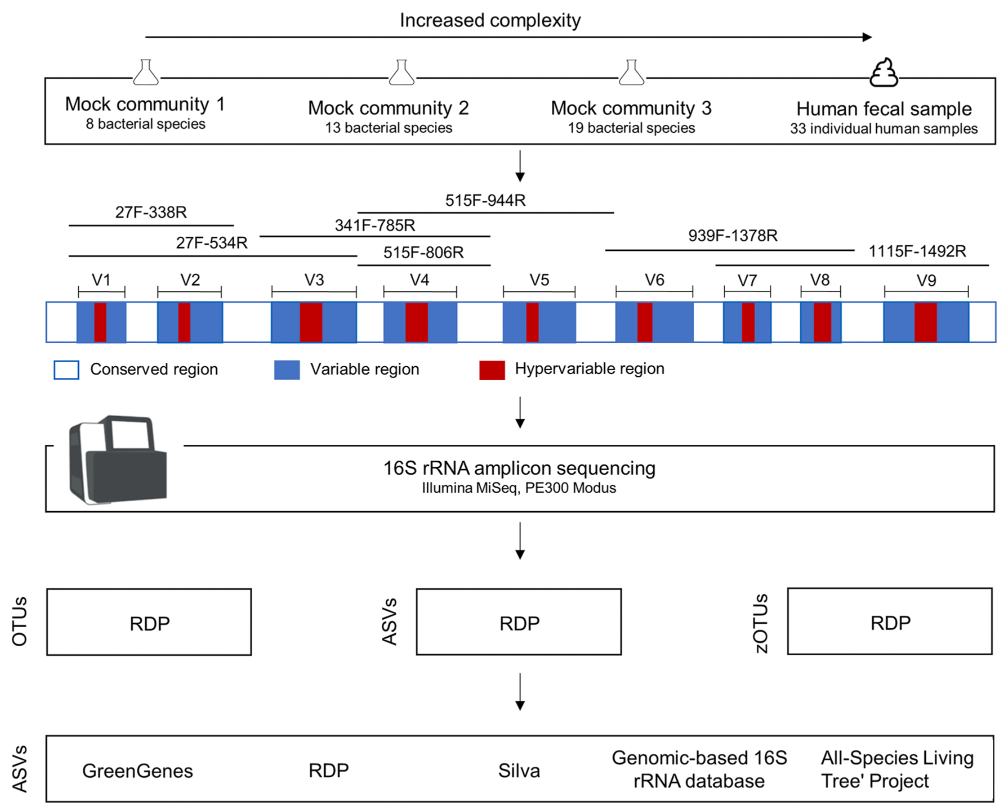
16S rRNA gene



(Fukuda et al., 2016, J UOEH)

- Key Features:
  - Highly conserved across all bacteria
  - Contains variable regions for species discrimination
  - ~1,500 bp length (9 hypervariable regions)
  - Slow evolutionary rate → stable for taxonomy

16S rRNA gene amplicon sequencing



(Abellan-Schneyder et al., 2021,mSphere)

- Typical Primers in Plant Root Microbiome Research

- 515F-806R
- 5′ GTGCCAGCMGCCGCGGTAA 3′
- 5′ GGACTACHVGGGTWTCTAAT 3′
- Amplification length ~250bp

- OTU (Operational Taxonomic Unit)

- ASV (Amplicon Sequence Variant)
- A Higher-Resolution Alternative to OTU Clustering

16S rRNA gene amplicon sequencing

1. Understanding Raw Sequencing Data

Assess sequencing quality using tools like FastQC or MultiQC.

2. Reads Alignment / Denoising

Clean and denoise reads (e.g., via DADA2, Deblur) or align to reference genomes.

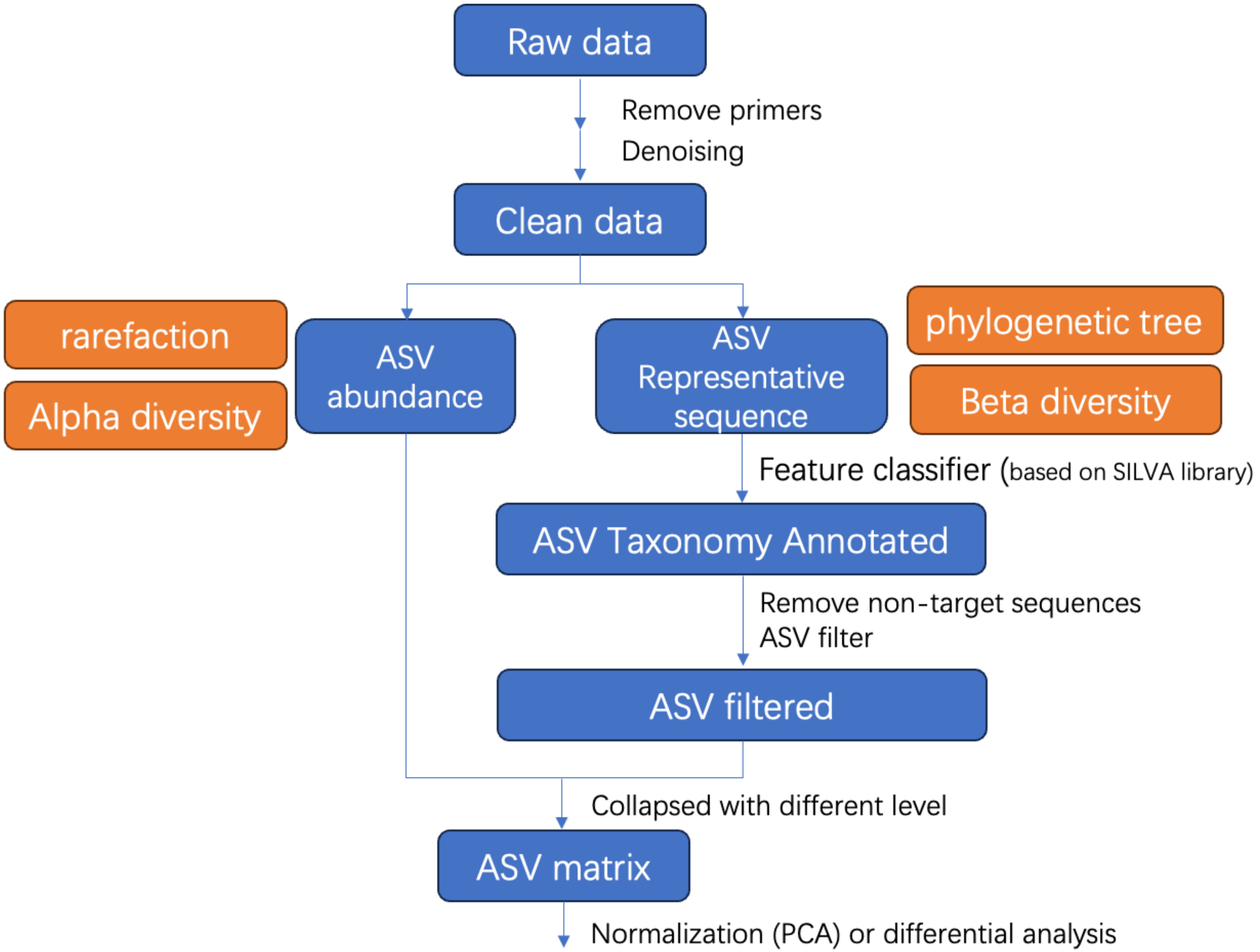
3. Taxonomic Classification

Assign taxonomy using reference databases (e.g., SILVA, Greengenes).

4. Quantification

Generate abundance tables for taxa or functional genes.

Workflow for 16S rRNA Sequencing Data Analysis (qiime)



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Preparation for the upcoming exercises

- References
- Test datasets
- Softwares

References

The ISME Journal (2021) 15:3181–3194
https://doi.org/10.1038/s41396-021-00993-z
ISME

ARTICLE

Check for updates

➢ Genome wide association study reveals plant loci controlling heritability of the rhizosphere microbiome

Siwen Deng[1,2] · Daniel F. Caddell[2] · Gen Xu[3,4] · Lindsay Dahlen[1,5] · Lorenzo Washington[1] · Jinliang Yang[3,4] · Devin Coleman-Derr[1,2]

Received: 27 May 2020 / Revised: 2 April 2021 / Accepted: 20 April 2021 / Published online: 12 May 2021
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nature plants
ARTICLES
https://doi.org/10.1038/s41477-021-00897-y

Check for updates

➢ Plant flavones enrich rhizosphere Oxalobacteraceae to improve maize performance under nitrogen deprivation

Peng Yu[1,2,3,17], Xiaoming He[1,2,3,17], Marcel Baer[2], Stien Beirinckx[4,5,6], Tian Tian[7], Yudelsy A. T. Moya[8], Xuechen Zhang[9], Marion Deichmann[10], Felix P. Frey[2], Verena Bresgen[2,3], Chunjian Li[11], Bahar S. Razavi[12], Gabriel Schaaf[10], Nicolaus von Wirén[8], Zhen Su[7], Marcel Bucher[13,14], Kenichi Tsuda[15,16], Sofie Goormachtig[4,6], Xinping Chen[1] ✉ and Frank Hochholdinger[1,2] ✉

Test datasets

Data type	Download link
16S sequencing raw data	SRR11839631 SRR11839632 SRR11839633 SRR11839634 SRR11839635 SRR11839636 SRR11839637 SRR11839639 SRR11839640
OTU matrix	https://github.com/colemanderr-lab/Deng-2020/blob/master/fig1_24line.rds
RNA-seq raw data	SRR11839278 SRR11839289 SRR11839301 SRR11839312 SRR11839497 SRR11839508 SRR11839619 SRR11839630 SRR11839780
Gene expression matrix	https://github.com/PengYuMaize/Yu2021NaturePlants/blob/master/gene_counts_table_WGCNA.txt
Reference genome	https://download.maizegdb.org/Genomes/B73/Zm-B73-REFERENCE-NAM-5.0/

Softwares

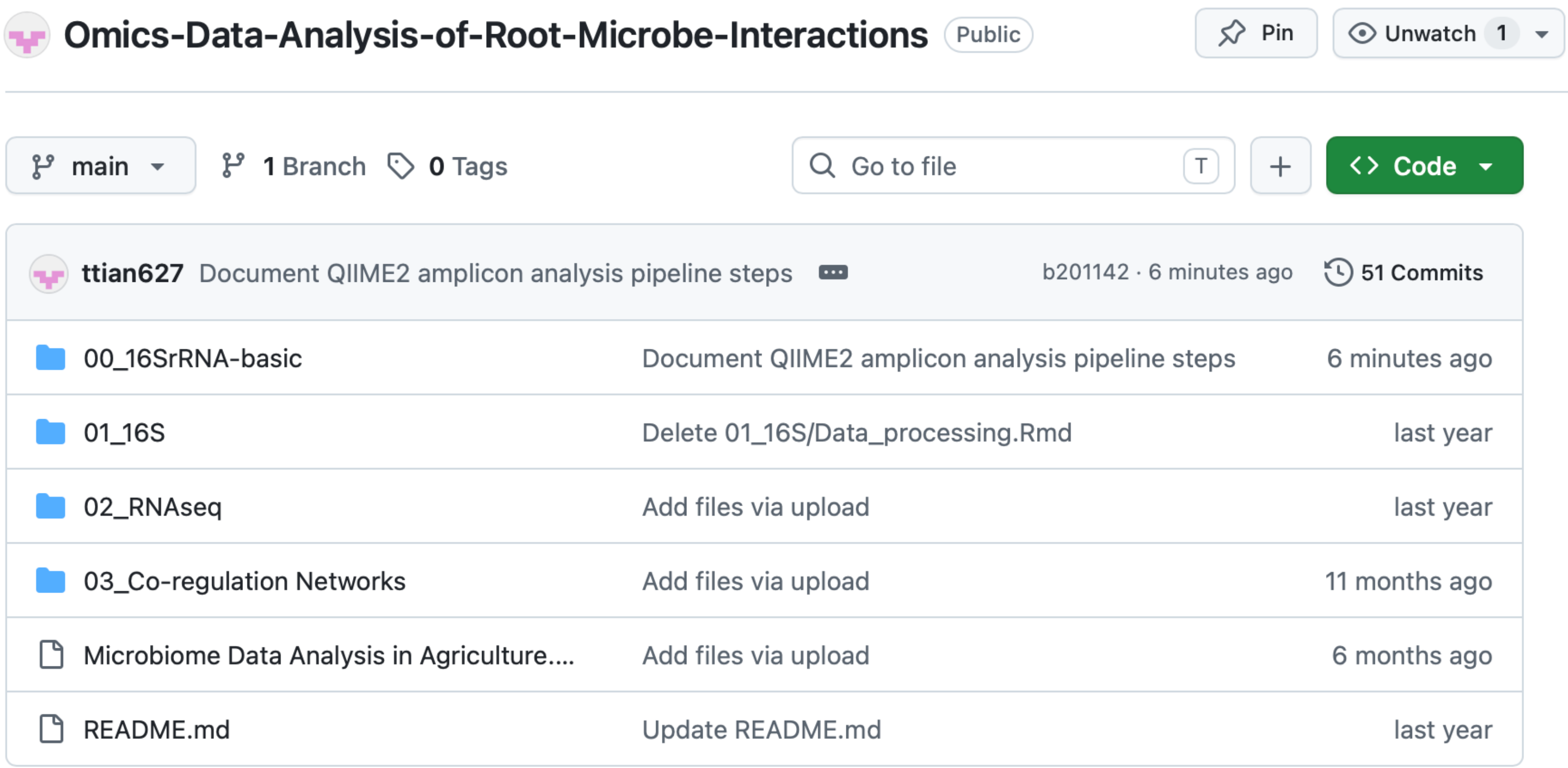
- Rstudio/R

- Conda
- sratoolkit (conda install -c bioconda sra-tools)
- hisat2
- stringtie
- samtools

- Install the appropriate version based on your system (Windows, macOS, or Linux).

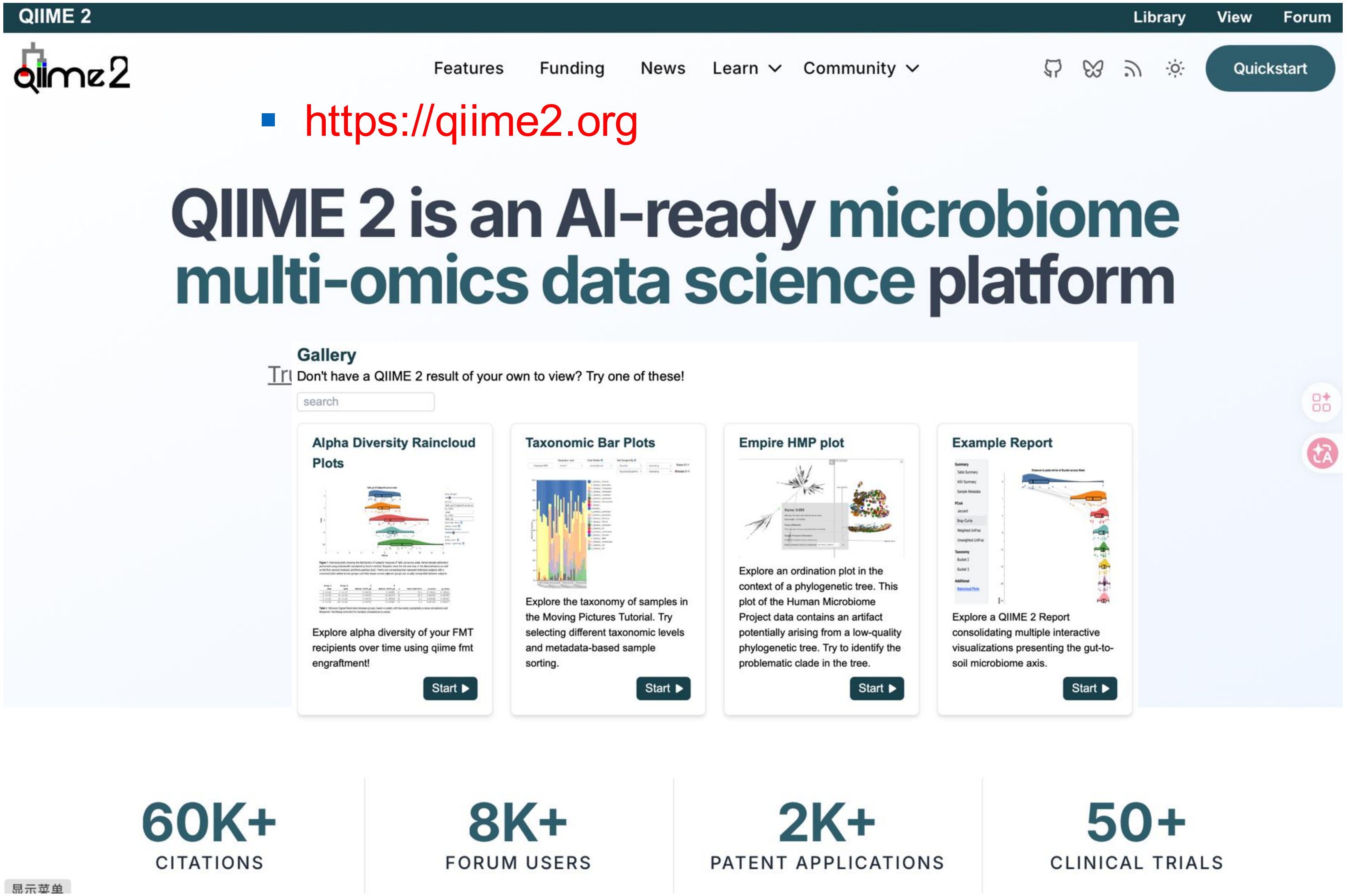
Download github info

https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions



Qiime2 website

- https://qiime2.org



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II Key Statistical Metrics in Microbiome Analysis

(27.5)

Key Statistical Metrics in Microbiome Analysis

- **Research objective:** To understand effective traits through the analysis of microbiome data.

  - **Alpha diversity**
  - **Rarefaction analysis**
  - **Beta diversity**
  - **Taxonomy distribution**

  - **Code:** https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/tree/main/01_16S

Key Statistical Metrics in Microbiome Analysis

➢ **Research objective:** To understand effective traits through the analysis of microbiome data.

- **Alpha diversity**
- **Rarefaction analysis**
- **Beta diversity**
- **Taxonomy distribution**

- **Code:** https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/tree/main/01_16S

Alpha diversity

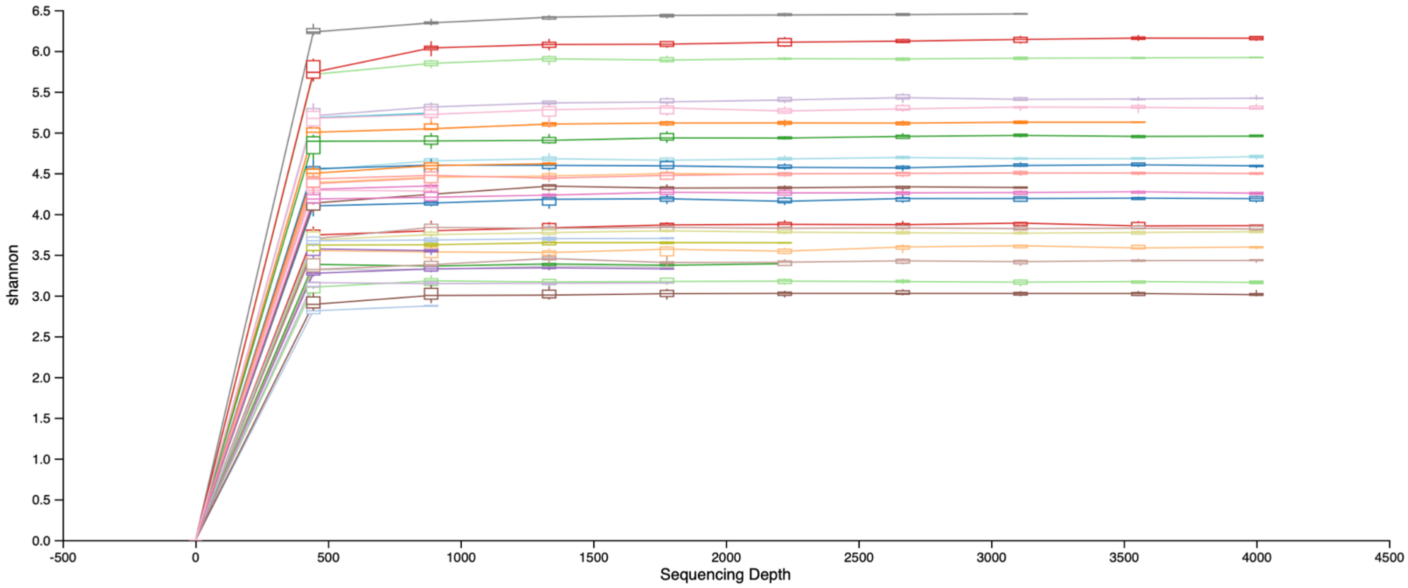
- Alpha diversity is within sample diversity.

- species diversity
- species evenness

Diversity Index	Description
Shannon Index	Considers both species richness and evenness; sensitive to rare species.
Simpson Index	Measures the probability that two randomly selected individuals belong to different species; more sensitive to dominant species.
Fisher's Alpha	Diversity index based on fitting a log series distribution; good for complex communities.

Rarefaction analysis

- Rarefaction curve illustrates how the cumulative number of alpha diversity metrics in a sample changes as sequencing depth increases.



From qiime2

Beta diversity

- Beta diversity is between sample diversity.
- It quantifies the dissimilarity or similarity in species composition between two or more samples.

Beta diversity	Species Composition	Species Abundance	Phylogenetic Tree
Jaccard index	✓		
Sorensen index	✓		
Bray-Curtis distance	✓	✓	
Euclidean distance	✓	✓	
Weighted UniFrac distance	✓	✓	✓

Taxonomy distribution

silva
high quality ribosomal RNA databases

GLOBAL CORE BIODATA RESOURCE

elixir Core Data Resource

de.NBI
GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

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SILVA

Welcome to the SILVA rRNA database project

A comprehensive on-line resource for quality checked and aligned ribosomal RNA sequence data.

SILVA provides comprehensive, quality checked and regularly updated datasets of aligned small (16S/18S, SSU) and large subunit (23S/28S, LSU) ribosomal RNA (rRNA) sequences for all three domains of life (***Bacteria***, ***Archaea*** and ***Eukarya***).

SILVA are the official databases of the software package **ARB**.

For more background information → Click here

SILVA Alignment, Classification and Tree (ACT) Service

The SILVA ACT service combines alignment, search and classify as well as reconstruction of trees in a single web application.

SILVA ACT is available at: → www.arb-silva.de/act

SILVAngs

silvangs

Check out our service for Next Generation Amplicon data

News

30.10.2024

DSMZ Digital Diversity Workshop

we are happy to announce that this year's DSMZ Digital Diversity Workshop will take place on 11 November 2024, 2 - 4 PM CET online!

11.07.2024

SILVA Release 138.2

The SILVA taxonomy release 138.2 is a maintenance release, with no additional sequences added to the ARB databases since the previous release 138.1. In this update we have focused on Prokaryotes.

11.12.2023

SILVA named Global Core Biodata Resource (GCBR)

SILVA is one of fifteen biodata resources worldwide that have been selected in the Global Biodata Coalition's 2023 selection round for Global Core Biodata Resources (GCBRs), increasing the total number of GCBRs to 52. SILVA joins its DSMZ sister databases BacDive and BRENDA (both selected in 2022) as well as LPSN (2023) as GCBR.

07.12.2023

We are hiring!

SILVA is looking for computer scientists as backend and web developer. Our colleagues from the DSMZ Digital Diversity integration team are also hiring for multiple positions. Find all open positions in our teams here.

go to Archive ->

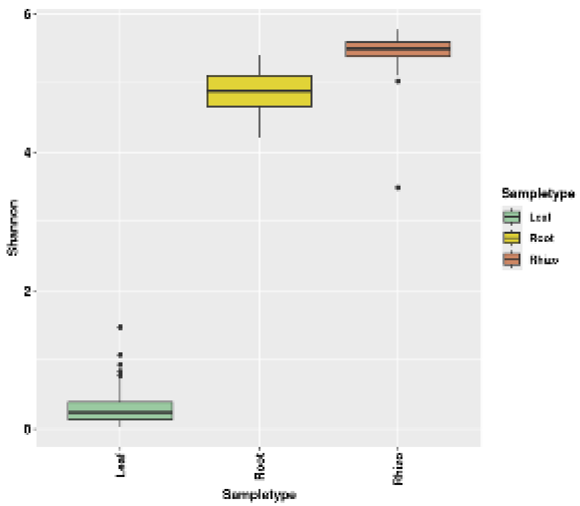
https://www.arb-silva.de

16S Data Structure

```
>rar <- readRDS("fig1_24line.rds")
> rar
phyloseq-class experiment-level object
otu_table() OTU Table: [ 1787 taxa and 204 samples ]
sample_data() Sample Data: [ 204 samples by 5 sample variables ]
tax_table() Taxonomy Table: [ 1787 taxa by 6 taxonomic ranks ]

>sample_data(rar)
>otu_table(rar)
>tax_table(rar)

> rar2 <- readRDS("fig3_200line.rds")
rar2
phyloseq-class experiment-level object
otu_table() OTU Table: [ 1189 taxa and 598 samples ]
sample_data() Sample Data: [ 598 samples by 16 sample variables ]
tax_table() Taxonomy Table: [ 1189 taxa by 7 taxonomic ranks ]
phy_tree() Phylogenetic Tree: [ 1189 tips and 1188 internal nodes ]
```

α diversity -shannon

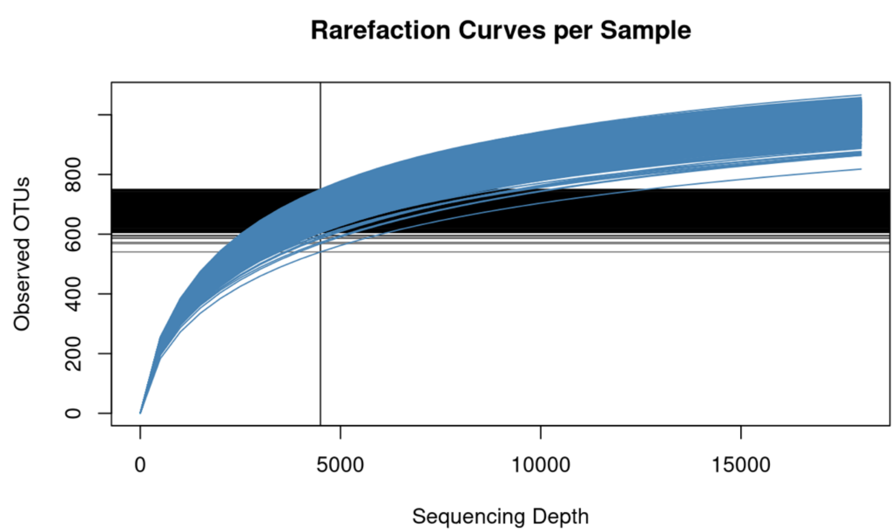
````
rar <- readRDS("fig1_24line.rds")
Diversity_table <- estimate_richness(rar,  measures=c("Shannon", "Observed"))
Diversity_table <- merge(Diversity_table, sample_data(rar), by="row.names")
Diversity_table$Sampletype <- factor(Diversity_table$Sampletype, levels = c( "Leaf", "Root", "Rhizo"))
```

Plot Figure 1b and Figure 1c
```{r}
p1b <- ggplot(data=Diversity_table, aes(y=Shannon,x=Sampletype,fill=Sampletype)) +
            geom_boxplot() +
            scale_fill_manual(values=c("#9ACAA1","#E1D337", "#CE8764")) +
            theme(axis.text.x=element_text(hjust=1,vjust=0.5,size=10,color="black",angle=90,face="bold"),
                  axis.text.y=element_text(size=11,color="black",face="bold"),
                  axis.title=element_text(size=11,face="bold"),text=element_text(size=11,face="bold"),
                  legend.position="right")
p1b
````



Deng et al., 2021

Rarefaction analysis

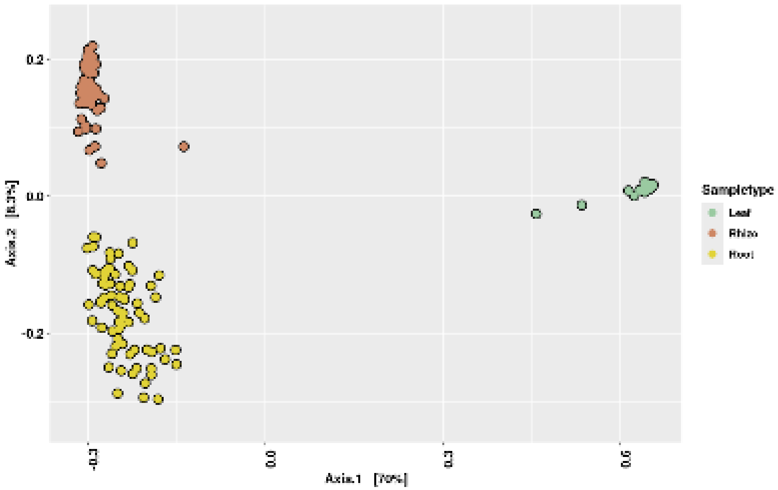
```r
rar1 <- readRDS("fig3_200line.rds")
otu_mat <- as(otu_table(rar1), "matrix")
if (taxa_are_rows(rar)) {
  otu_mat <- t(otu_mat)
}

rarecurve(otu_mat,
          step = 500,
          sample = depth,
          col = "steelblue",
          cex = 0.5,
          label = FALSE,
          ylab = "Observed OTUs",
          xlab = "Sequencing Depth",
          main = "Rarefaction Curves per Sample")
```



Deng et al., 2021

β diversity - Bray-Curtis distance

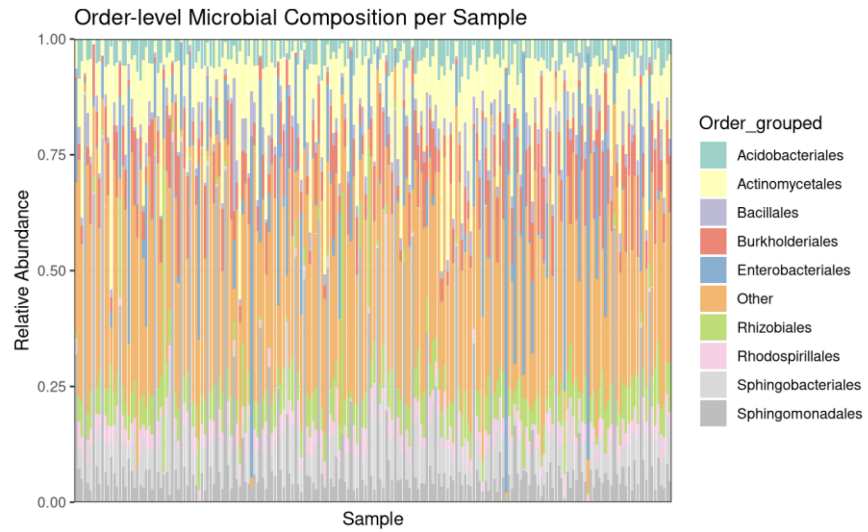
```
p1c <- plot_ordination(rar, ordinate(rar, "MDS",distance="bray"),axes=1:2, color = "Sampletype") +
                              scale_color_manual(values=c("#9ACAA1","#CE8764","#E1D337")) +
                              geom_point(colour="black",size=3.5) +
                              geom_point(size = 2.5) +
                              xlim(-0.4,0.7) +
                              scale_x_continuous(breaks=c(-0.3,0,0.3,0.6))+
                              ylim(-0.33,0.25)+
                              theme(axis.text.x=element_text(size=11,color="black",angle=90),
                                    axis.text.y=element_text(size=11,color="black"),
                                    axis.title=element_text(size=11,face="bold"),
                                    text=element_text(size=11,face="bold"),
                                    legend.position="right")
p1c
```



Deng et al., 2021

Taxonomy distribution

```r
rar_order <- tax_glom(rar, taxrank = "Order")
# compute Order abundance for each sample
rar_order_rel <- transform_sample_counts(rar_order, function(x) x / sum(x))
# change phyloseq to dataframe
df_order <- psmelt(rar_order_rel)
# set NA as "Unclassified"

# top10 Orders and "Other"
top_orders <- df_order %>%
  group_by(Order) %>%
  summarise(Abundance = sum(Abundance)) %>%
  top_n(9, Abundance) %>%
  pull(Order)

df_order$Order_grouped <- ifelse(df_order$Order %in% top_orders, df_order$Order, "Other")

# plot
ggplot(df_order, aes(x = Sample, y = Abundance, fill = Order_grouped)) +
  geom_bar(stat = "identity") +
  theme_bw() +
  labs(title = "Order-level Microbial Composition per Sample",
       x = "Sample",
       y = "Relative Abundance") +
  theme(axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        legend.position = "right") +
        scale_y_continuous(limits = c(0, 1), expand = c(0, 0))+
  scale_fill_manual(values = c(RColorBrewer::brewer.pal(9, "Set3"), "grey"))
```



Deng et al., 2021

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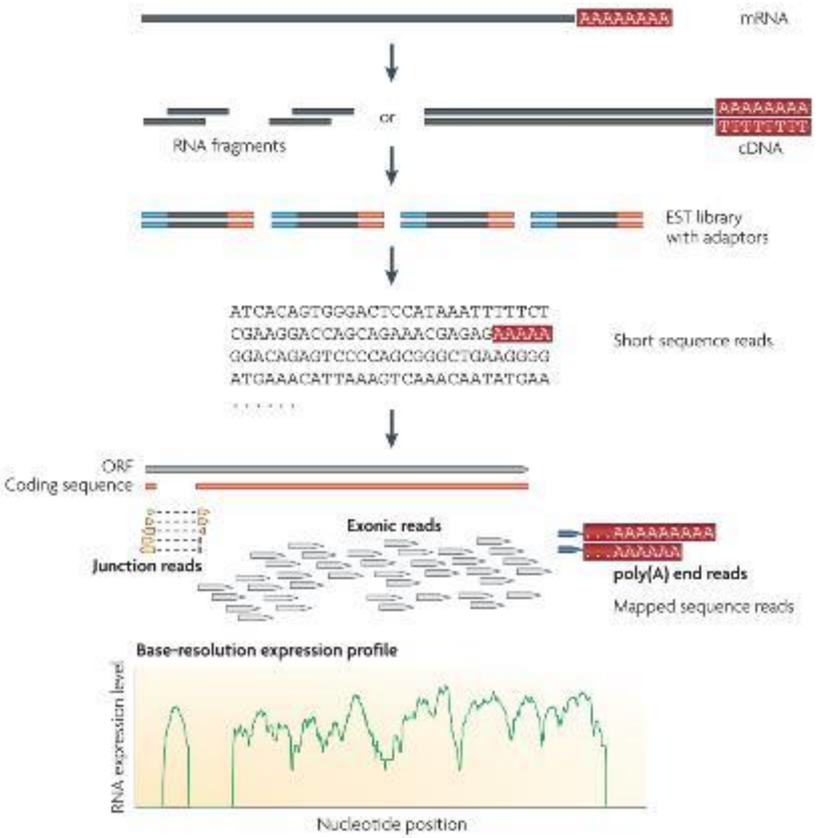
III Root transcriptome data analysis (3.6)

Transcriptome Sequencing

- **Research objective:** To understand the standard RNA-seq data analysis workflow and construct an expression matrix.



(Wang et al.,2009, Nature Reviews Genetics)

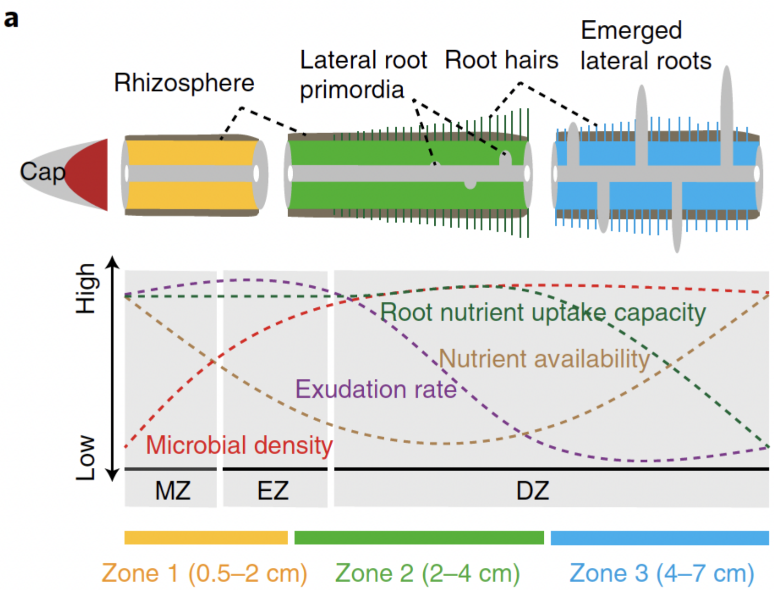
- Core Applications of RNA-seq
  - Quantifying gene expression levels
  - Detecting alternative splicing events
  - Assembling transcripts

(de novo or reference-guided)

  - Detecting gene fusions

Sample information

30 Maize inbred lines



(Yu et al.,2021, Nature Plants)

Run	Sample name	Description
SRR11839780	B73-Z1-1	zone1 of crown root
SRR11839312	B73-Z1-2	zone1 of crown root
SRR11839301	B73-Z1-3	zone1 of crown root
SRR11839289	B73-Z2-1	zone2 of crown root
SRR11839278	B73-Z2-2	zone2 of crown root
SRR11839508	B73-Z2-3	zone2 of crown root
SRR11839497	B73-Z3-1	zone3 of crown root
SRR11839630	B73-Z3-2	zone3 of crown root
SRR11839619	B73-Z3-3	zone3 of crown root

LH38
LH39
A9
787
IB014
PHN11
PHJ75
PHM10
LH205
PHW51
B73
A632
Mo17
LH51
LH52
E8501
LH93
PHJ31
PHW20
W64A

RNA sequencing data analysis pipeline

1. Understanding Raw Sequencing Data
Check read quality using FastQC or MultiQC.

2. Read Trimming & Alignment / Quantification
Trim adapters and low-quality reads, then align reads to a genome (STAR, HISAT2).

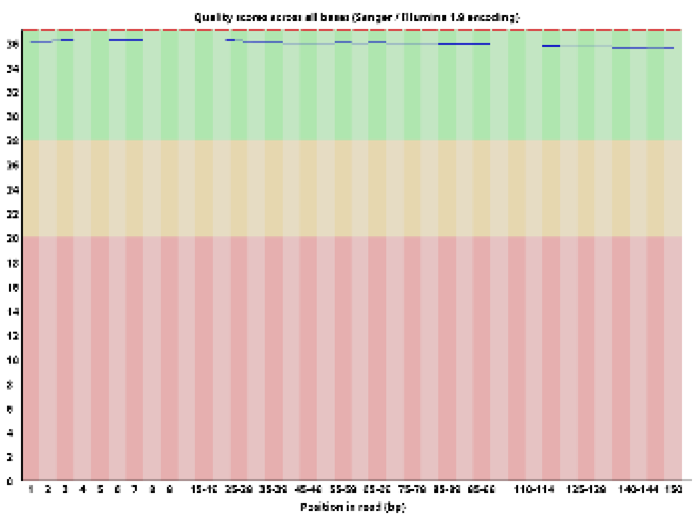
3. Gene / Transcript Quantification
Generate expression matrices (e.g., counts, TPM, FPKM).

4. Data Visualization & Interpretation
Visualize patterns using PCA, heatmaps, volcano plots, etc.

1. Understanding Raw Sequencing Data

Raw reads

```
[go82tip@tuwzz7x-odin RNAseq]$ head SRR11839278_1.fastq
@SRR11839278.1 1 length=150
TCTCACCCTTGTCTCGCGCAACAGCAGTCGGCCGCGGGCCCGGGATCGCTGGCGGGTTGTGCGGGGTTGTGCTCACCGCTGCTACTAGCC
GAGCGAGAGAGAGAGAGAGAAG
+SRR11839278.1 1 length=150
FF:F,FFFF,F,F:FFFFFF,F:FF,F:FFF,F,,FFFFFF,,F,:FFFFFFFFFFF,FFFFF,FF,,F,FF:F:FFFFFFF:,F,,FF,
F,F,F,F:FFF,F:FFF:FF,F
@SRR11839278.2 2 length=150
CAGGTCCTAAAATATGGTTTCTCGCAATCAAATACTTCATTCTTGGATGCCCCGGCGCCTACTACCTTTGGTCCCGGCCACTATACCGTG
TTCCTATTCTACTTGGTTCACA
+SRR11839278.2 2 length=150
FFFFFFF::FFFFFFF:FF,FFFFFF:,F,,FFFFF,F:::FF,FFFFFFFFFF,FFFFFFFF,FF:F::,:,FFFFFF:F:FFFFFF:F
FFFFF,FFFFFFFF:,::FFFF
```



2. Alignment / Quantification

Bam file

```
(bioenv) [go82tip@tuwzz7x-odin RNAseq]$ samtools view SRR11839278.uniq.bam | less -S
SRR11839278.15888007    99    chr1    21632    60    18S9M22508N123M =       45636    375     GGAGACCTGGAGGAGACGCTGGAGGGG
SRR11839278.9766371     113   chr1    34648    60    150M   =       34793    275     CGGGAACGGAACAAAGACACGCGGCGTGCGGGGCC
SRR11839278.23144857    163   chr1    34659    60    150M   =       34850    341     CAAAGACACGCGGCGTGCGGGGCCTCGCTCCTCTG
SRR11839278.1506825     99    chr1    34667    60    150M   =       34750    233     CGCGGCGTGCGGGGCCTCGCTCCTCTGGTCTTCCC
```

position 150 match

Mapping result

```
[go82tip@tuwzz7x-odin RNAseq]$ more SRR11839278.summary
26469714 reads; of these:
  26469714 (100.00%) were paired; of these:
    2116811 (8.00%) aligned concordantly 0 times
    23712176 (89.58%) aligned concordantly exactly 1 time
    640727 (2.42%) aligned concordantly >1 times
    ----
    2116811 pairs aligned concordantly 0 times; of these:
      163073 (7.70%) aligned discordantly 1 time
    ----
    1953738 pairs aligned 0 times concordantly or discordantly; of these:
      3907476 mates make up the pairs; of these:
        2389423 (61.15%) aligned 0 times
        1473075 (37.70%) aligned exactly 1 time
        44978 (1.15%) aligned >1 times
95.49% overall alignment rate
```

3. Gene / Transcript Quantification

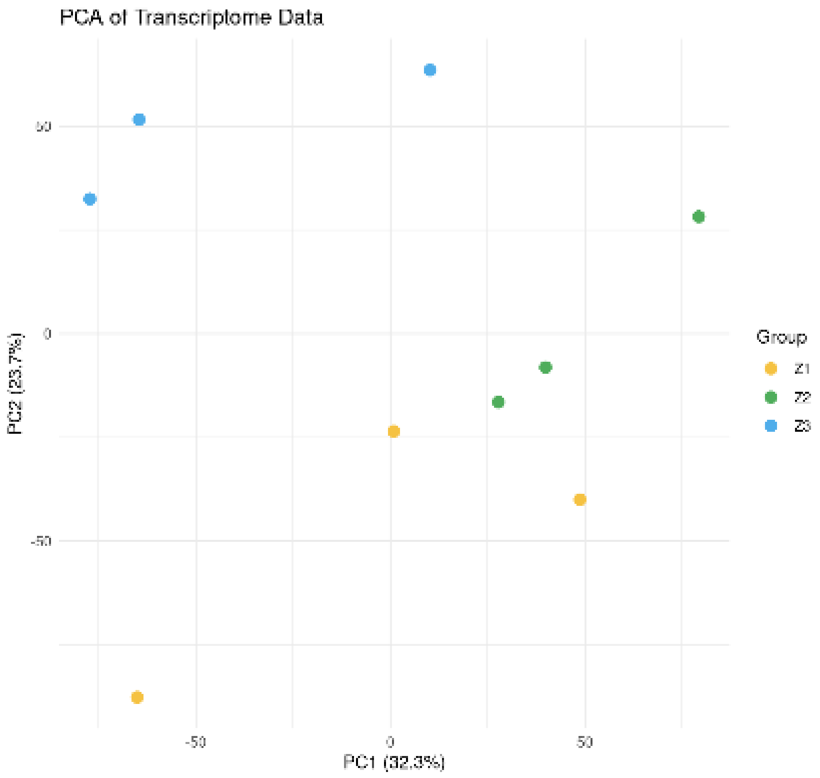
```
(bioenv) [go82tip@tuwzz7x-odin RNAseq]$ head stringtie_uniq/B73-Z2-2/SRR11839278.abun
Gene ID Gene Name       Reference       Strand  Start   End     Coverage        FPKM    TPM
Zm00001eb000150 -       chr1    +       328087  328525  1.637813        0.241440        0.423117
Zm00001eb000170 -       chr1    -       550834  564766  34.270252       5.897928        10.335961
Zm00001eb000180 -       chr1    +       551078  552061  0.000000        0.000000        0.000000
Zm00001eb000010 -       chr1    +       34617   40204   9.302351        1.371314        2.403192
Zm00001eb000020 -       chr1    -       41214   46762   117.030136      22.718590       39.813721
Zm00001eb000050 -       chr1    -       108554  114382  0.000000        0.000000        0.000000
Zm00001eb000060 -       chr1    -       188559  189581  113.641823      16.752607       29.358500
Zm00001eb000070 -       chr1    -       190192  198832  0.143411        0.021141        0.037049
Zm00001eb000080 -       chr1    -       200262  203393  128.814728      18.989334       33.278301
```

FPKM（Fragments Per Kilobase of transcript per Million mapped reads）

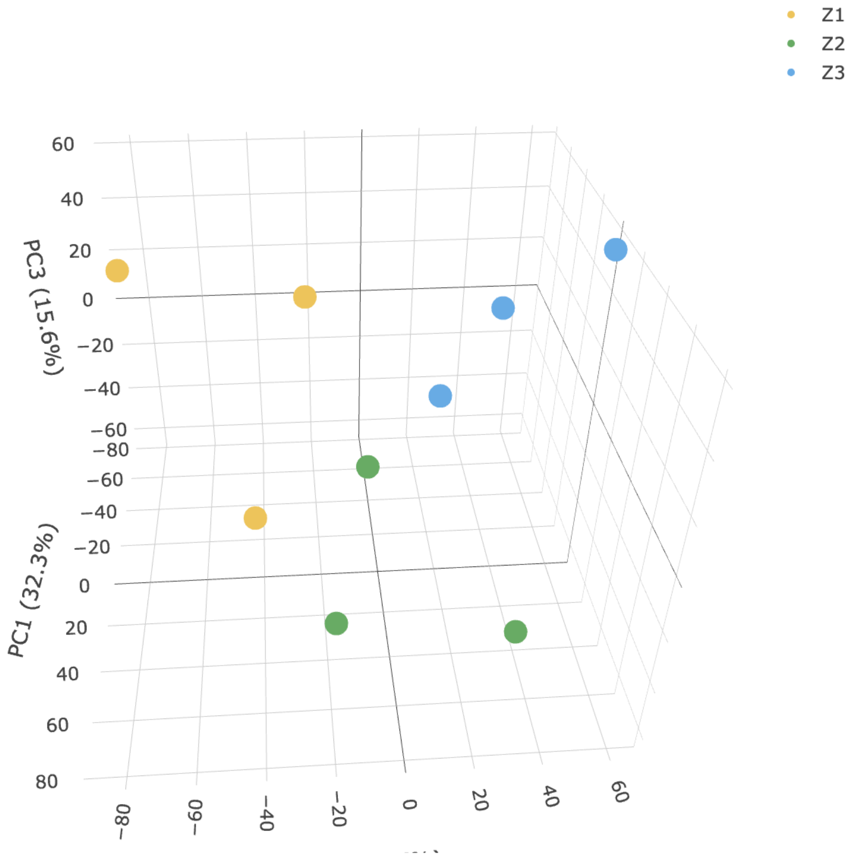
TPM（Transcripts Per Million）

4. Data Visualization & Interpretation

2D-PCA



3D-PCA



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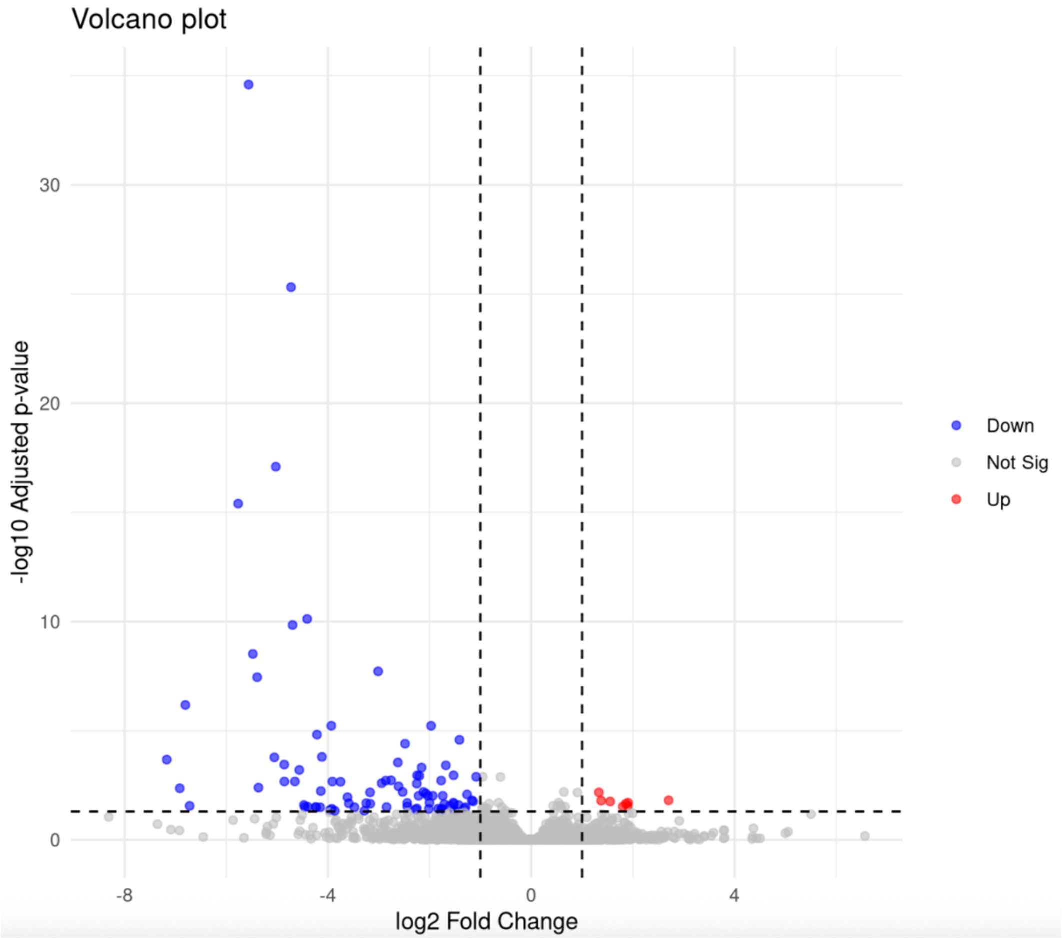
4. Data Visualization & Interpretation

Differential expression analysis



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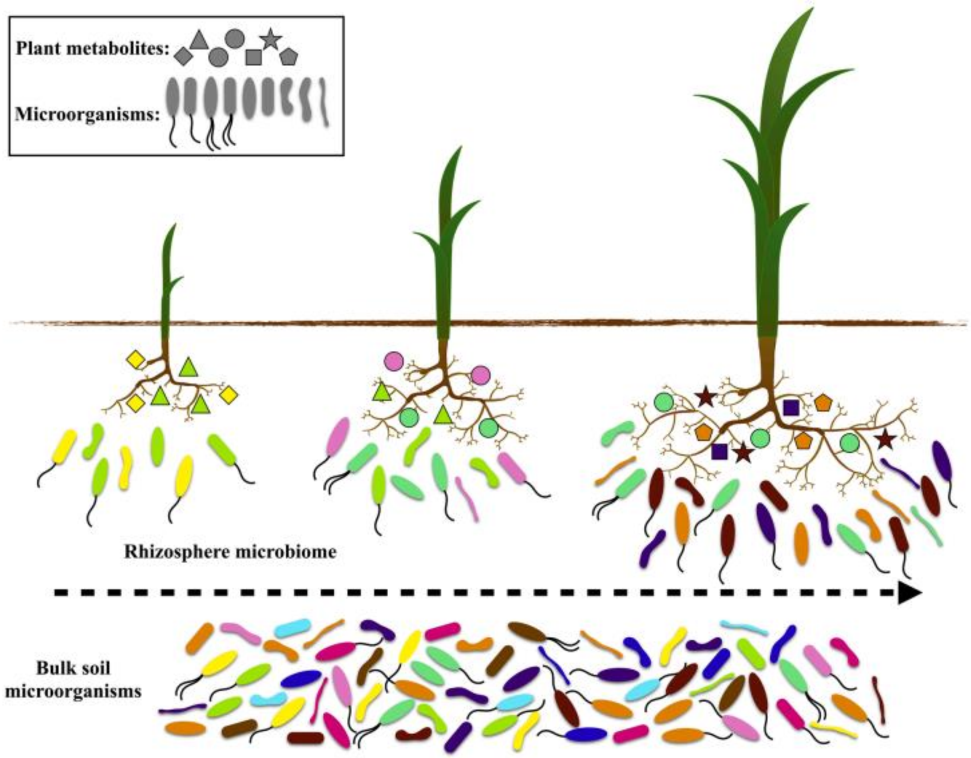
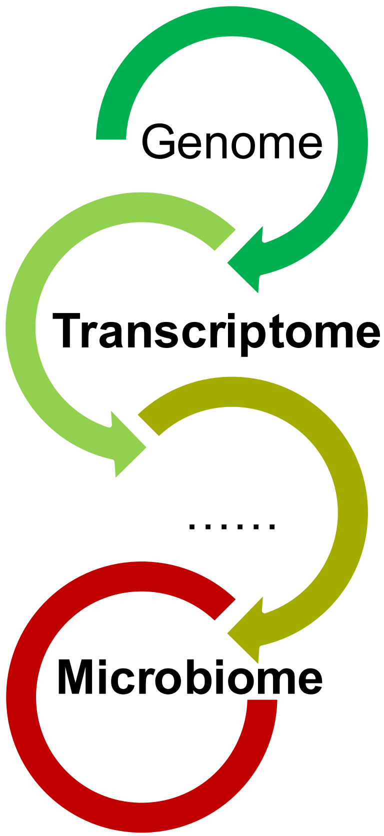
IV Gene Expression – Microbiome Co-regulation Networks (10.6)

Root and Microbiome interactions

- **Research objective:** To identify the regulatory relationships between specific gene expression and the abundance of specific microbes.



(Zhang et al., 2021, Microbiological Research)



Microbiome co-regulation Networks

- Expression matrix

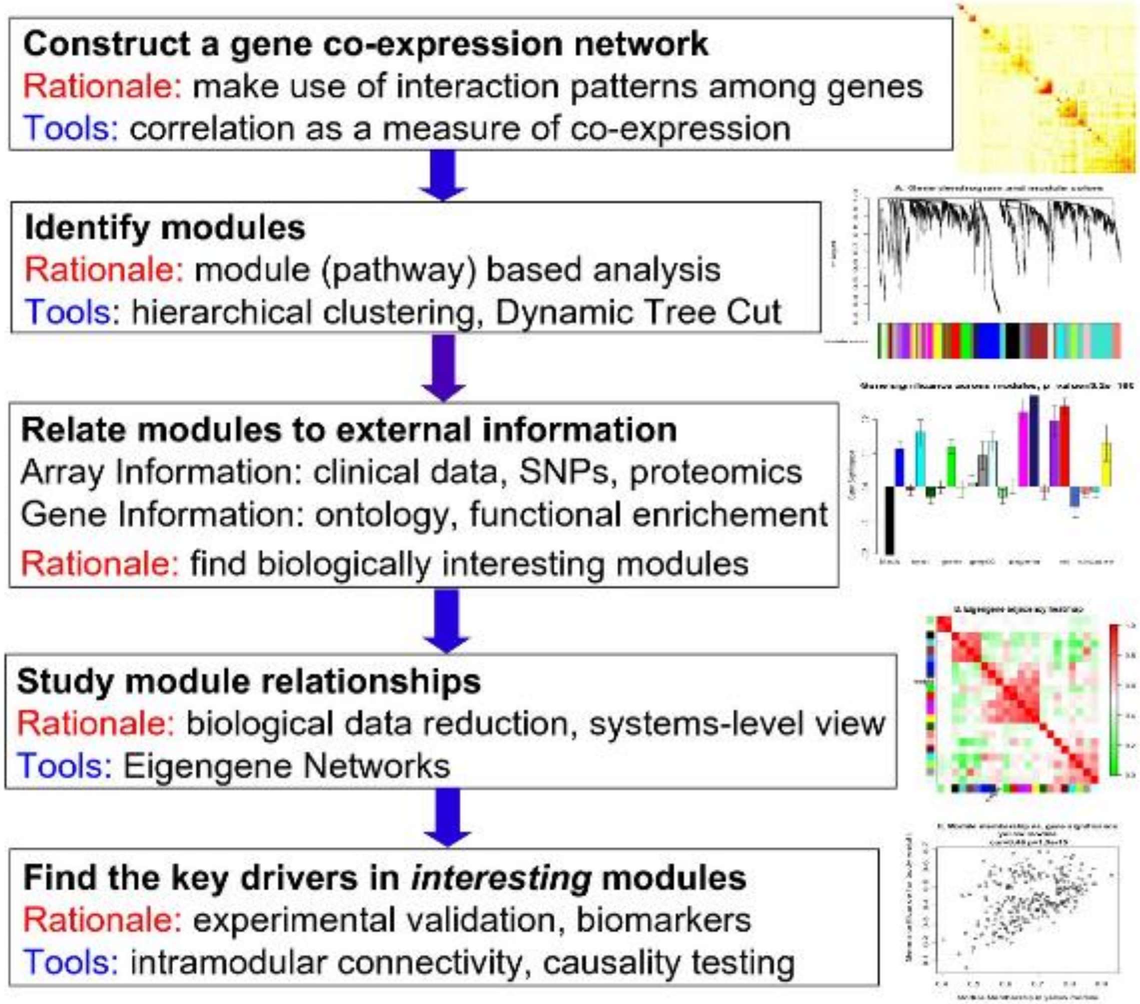
	sample1	sample2
Gene1	FPKM1	FPKM2
Gene2	FPKM3	FPKM4
Gene3	FPKM5	FPKM6
…	…	…

- Microbiome ASV feature table

	sample1	sample2
class1	count1	count2
class2	count3	count4
class3	count5	count6
…	…	…

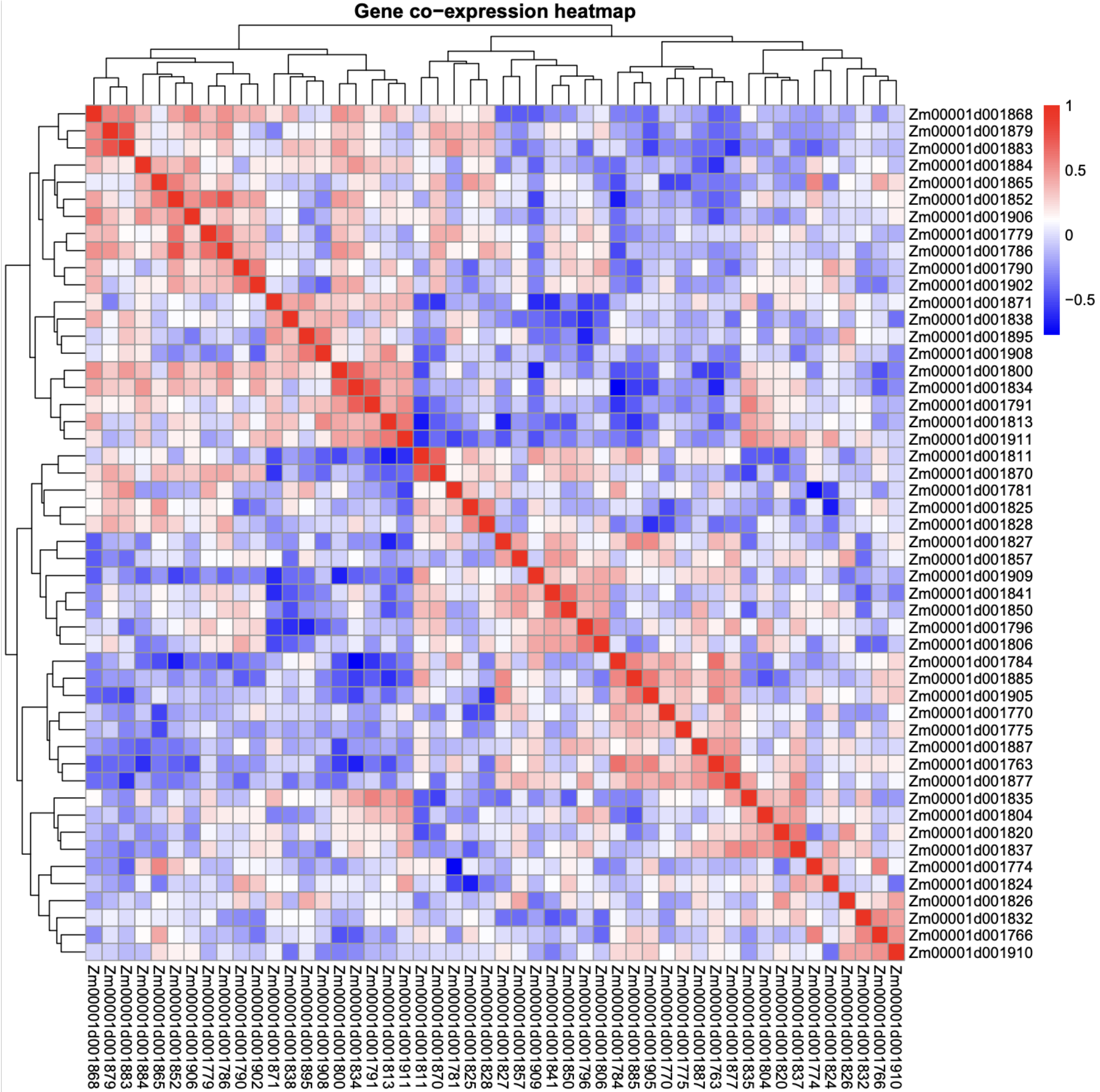
- Code
- https://github.com/ttian627/Omics-Data-Analysis-of-Root-Microbe-Interactions/03_Co-regulation Networks/

Microbiome Co-regulation Networks

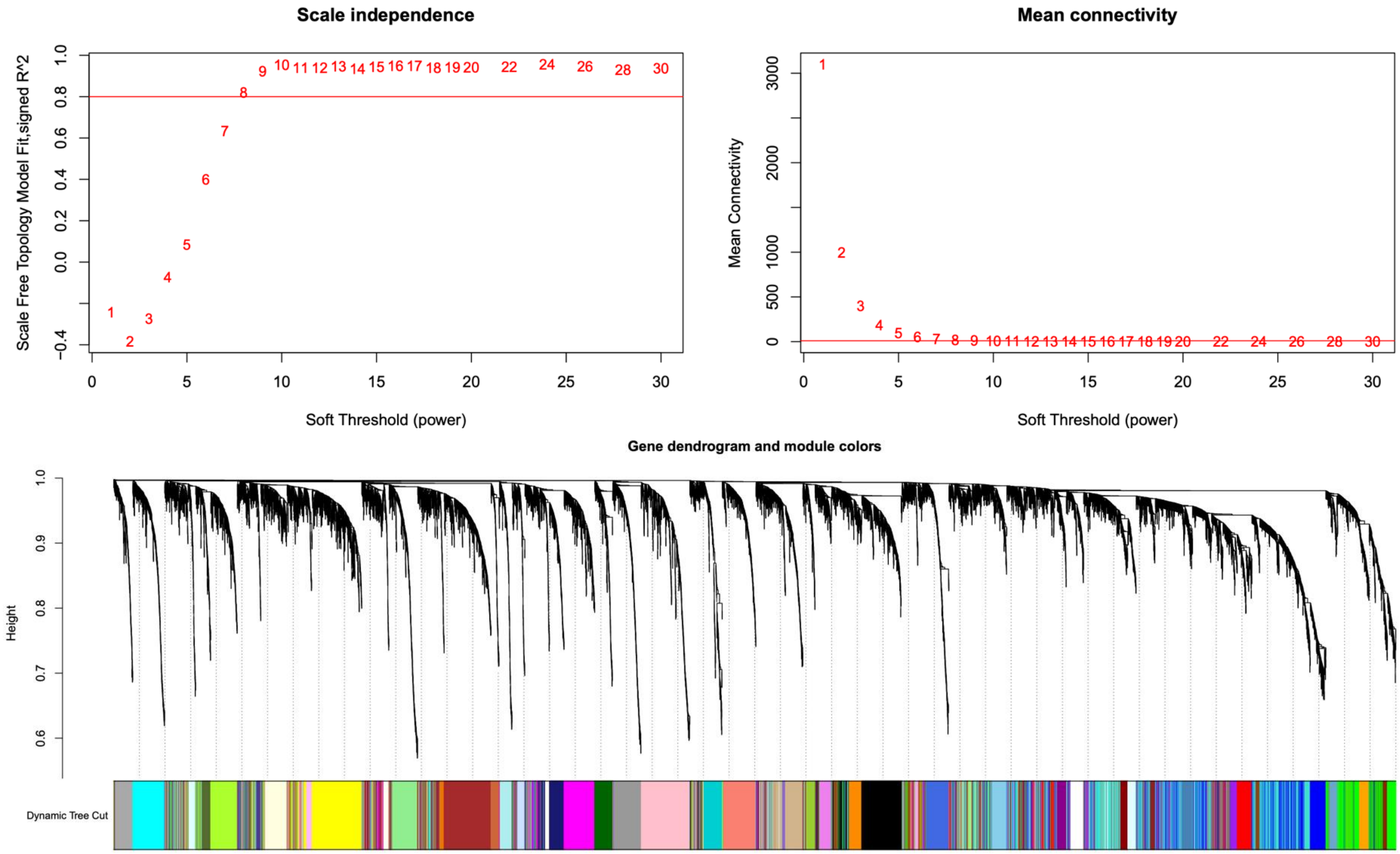


(Peter Langfelder, et al., 2008, Genome Biology)

Construct a gene co-expression network



Identify modules



How many gene modules have you obtained?

Relate modules to external information

Please use one of the gene modules on the following website to perform the functional enrichment analysis.

Select on gene module (less than 100 genes)

Convert the gene ID from v4 to v5

Based on the convet file from github

B73v4_to_B73v5.tsv

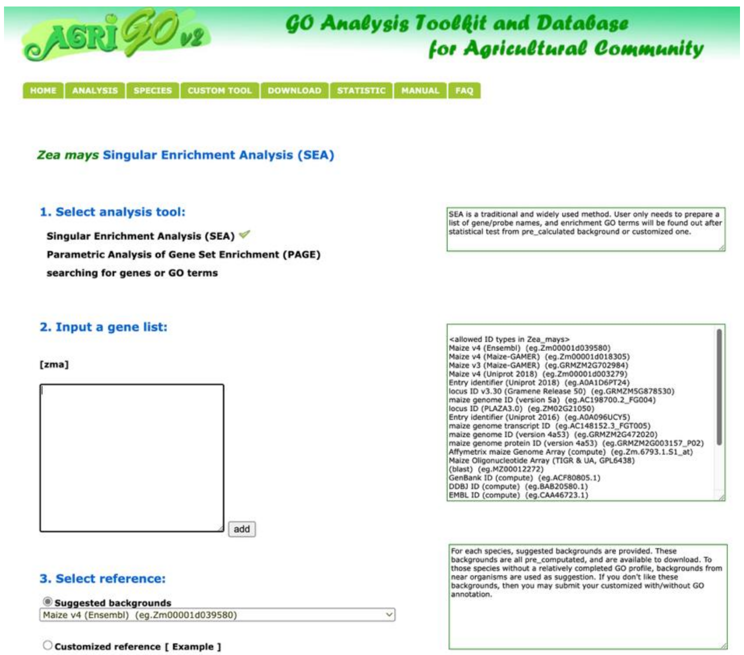
Conduct GO enrichment analysis

https://v2.arabidopsis.org/tools/go_term_enrichment.jsp

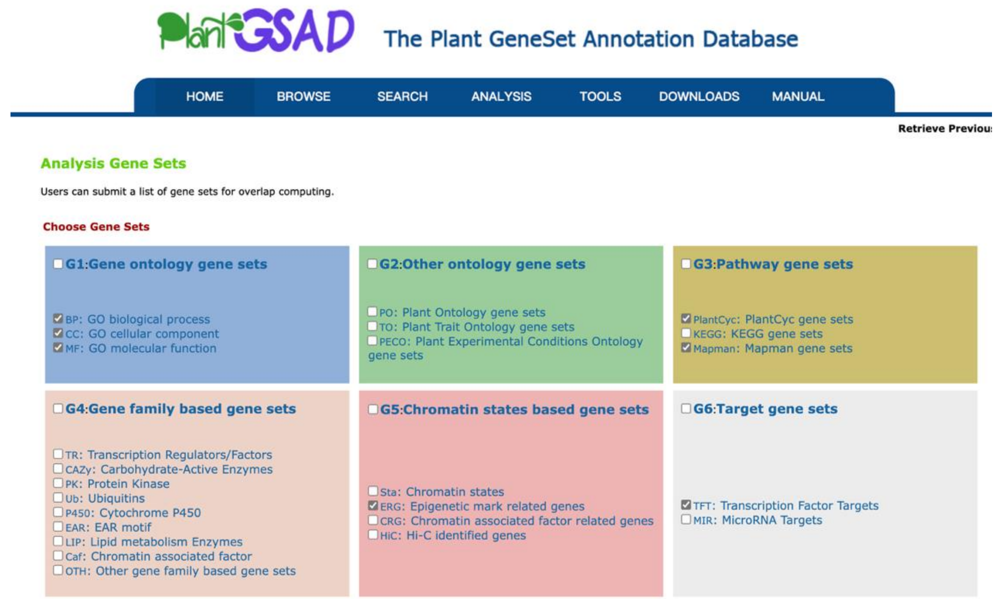
Relate modules to external information

Please use one of the gene modules on the following website to perform the functional enrichment analysis.

agriGO



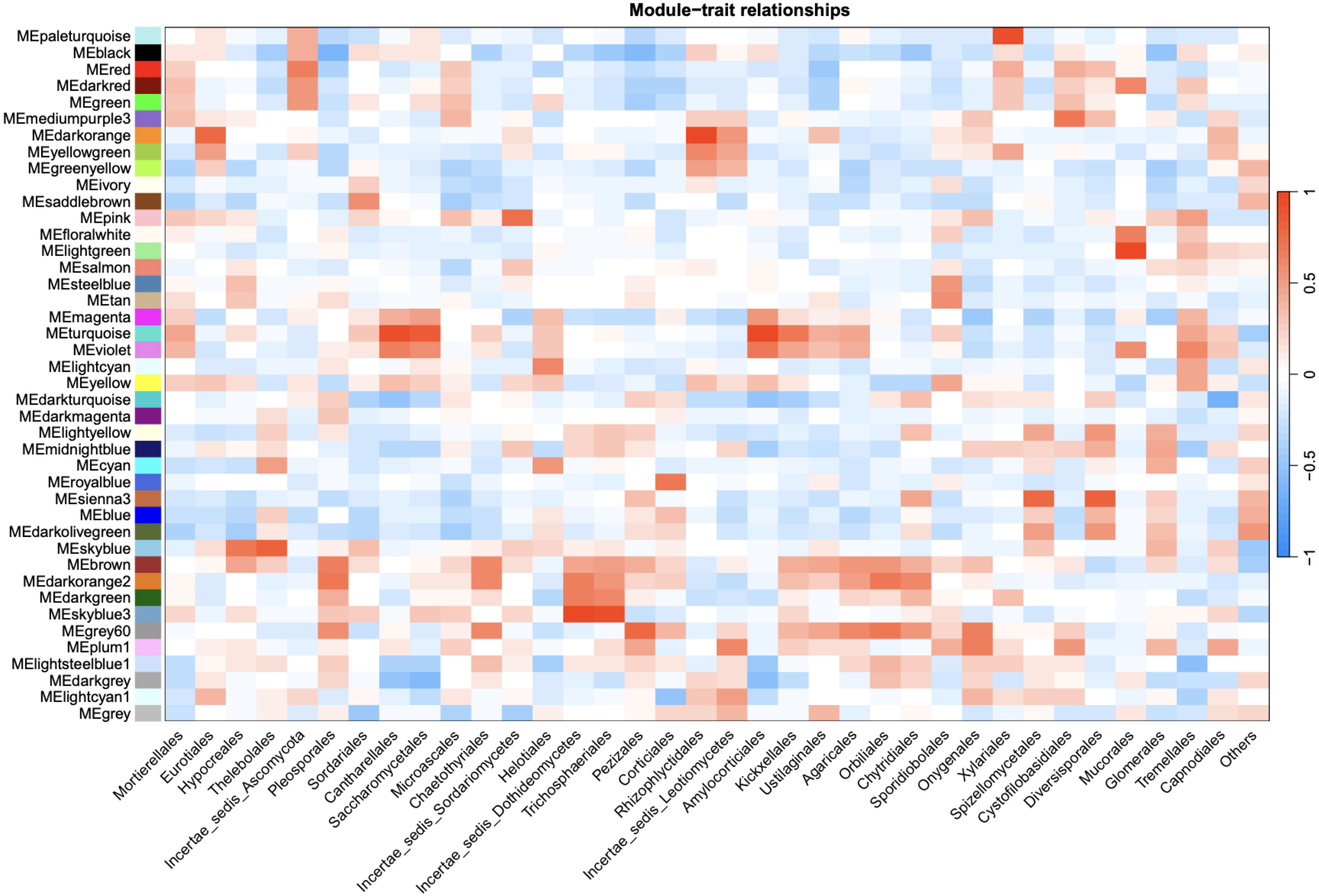
PlantGSAD



https://systemsbiology.cau.edu.cn/agriGOv2/specises_analysis.php?SpeciseID=20&latin=Zea_mays

https://systemsbiology.cau.edu.cn/PlantGSEAv2/analysis.php

Microbiome Co-regulation Networks



- Could you try constructing co-expression networks using different transcriptome datasets or alternative microbial classifications?

S1 safety instruction !!!

Thank you

Lecturer Assistant
Tian Tian